Fair and Accurate Skin Disease Image Classification using Alignment with Clinical Text Representations

Aayushman [2] Hemanth Gaddey [1] Vidhi Mittal [1] Manisha Chawla [1] Gagan Raj Gupta [1]

[1]Indian Institute of Technology Bhilai [2]Indian Institute of Science Education and Research Bhopal

Motivation





Problem Statement

Can we design an accurate skin disease classification model using small datasets like DDI or noisy datasets like Fitzpatrick? How can we improve Fairness?



Figure 1. Fitzpatrick Scale

Figure 2. Class-Label Distribution of Benign and Malignant classes in FitzPatrick17k Dataset

Data imbalance across skin tones as per Fitzpatrick scale presents additional hurdle in fair classification.

Take away

The key idea is to leverage the information and relationships in the clinical text embeddings and improve the image embeddings through the alignment loss.

Proposed Approach for Alignment

$$\mathcal{L}_{\text{GOT}}(\boldsymbol{\mu}, \boldsymbol{\nu}, M) = \min_{\mathbf{T} \in \Pi(\mathbf{u}, \mathbf{v})} \sum_{i,i',j,j'} M \odot \mathbf{T}_{ij} \Big(\lambda c(x_i, y_j) + (1-\lambda) M \odot \mathbf{T}_{i'j'} c(x_i, y_j, x'_i, y'_j) \Big) \quad (1)$$

With $(\mathbf{M} \odot \mathbf{T})\mathbf{1}_m = u$, $(\mathbf{M} \odot \mathbf{T})^T \mathbf{1}_n = v$ and $\mathbf{T} \odot (1_{n \times m} - M) = 0_{n \times m}$ where $\mathbf{1}^m$ is the m-dimensional vector of ones.

The optimal transport plan $\mathbf{T}$ is obtained by minimizing the $\mathcal{L}_{GOT}$ using the iterative Sinkhorn algorithm to solve (1) with an entropic regularizer.

Clinical Text Representations

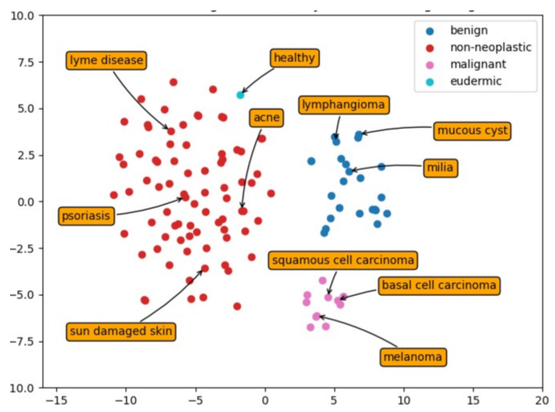


Figure 3. t-SNE Plot of Embeddings Generated text-embedding-3-large model

Model Architecture

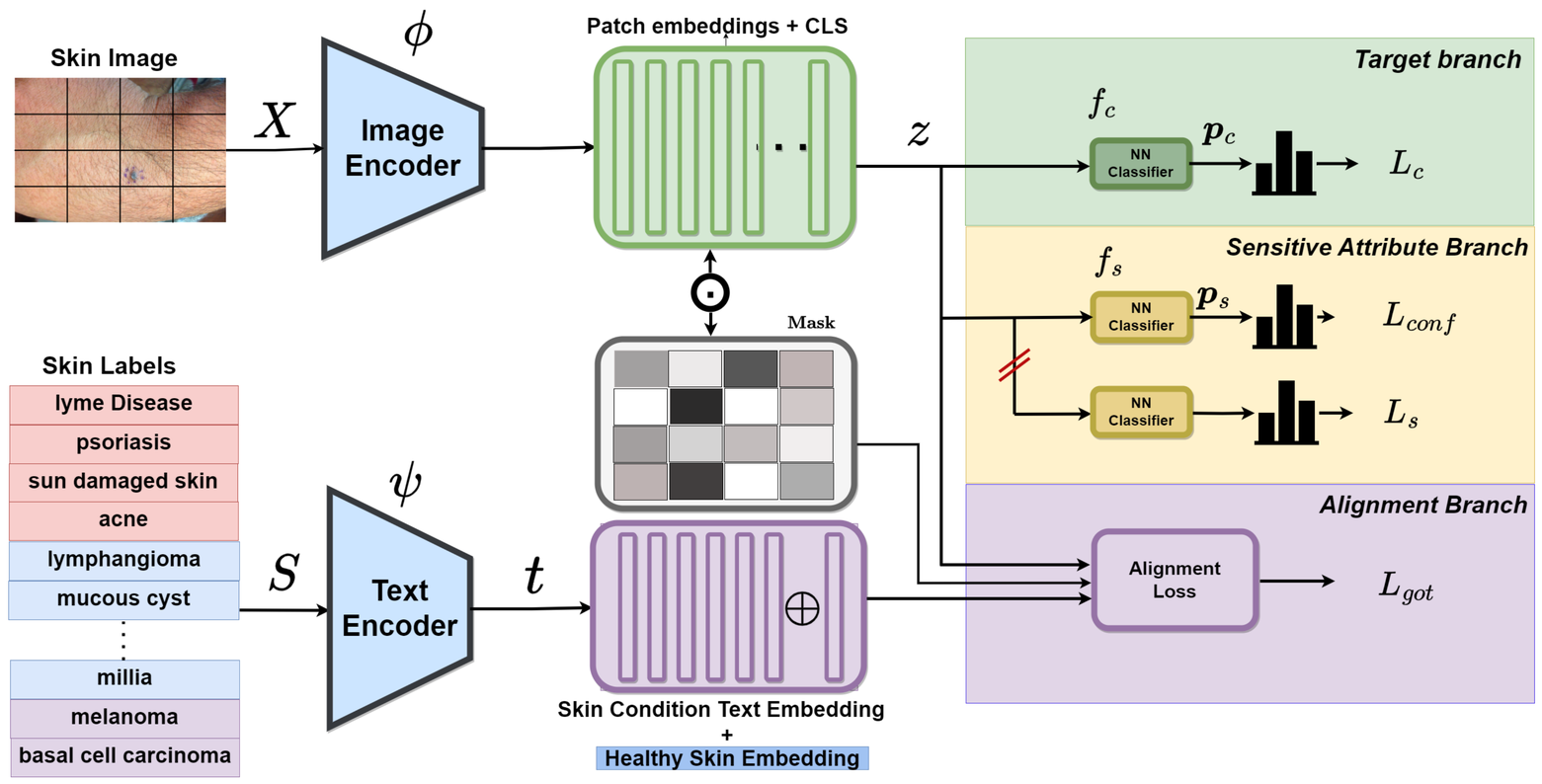


Figure 4. PatchAlign Model Architecture

- Sensitive Attribute Branch is used to suppress skin tone features from text embedding.
- Alignment with Skin Disease Label to enhance features of patches. These labels have no bias of skin type.
- Eudermic (healthy) skin is used to align disease free patches.
- Image Encoder converts the image into patches and produces patch representation with global image representation

Fairness (%)

1. **Predictive Quality Disparity (PQD):** Measures the consistency in model accuracy across different skin types.

$$\text{PQD} = \frac{\min(acc_j, j \in S)}{\max(acc_j, j \in S)} \quad (2)$$

2. **Equality of Opportunity Measure (EOM):** Evaluates fairness by comparing the true positive rates across groups, aiming for equal opportunity regardless of skin type.

$$\text{EOM} = \frac{1}{M} \sum_{i=1}^{M} \frac{\min[p(\hat{y} = i | y = i, s = j), j \in S]}{\max[p(\hat{y} = i | y = i, s = j), j \in S]} \quad (3)$$

Where S is the set of skin types while M is the set of skin conditions.

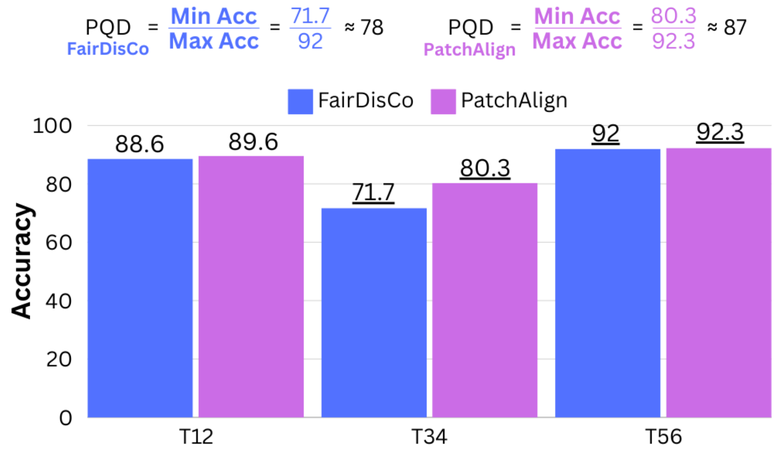


Figure 5. PQD calculated from DDI dataset across 3 skin types.

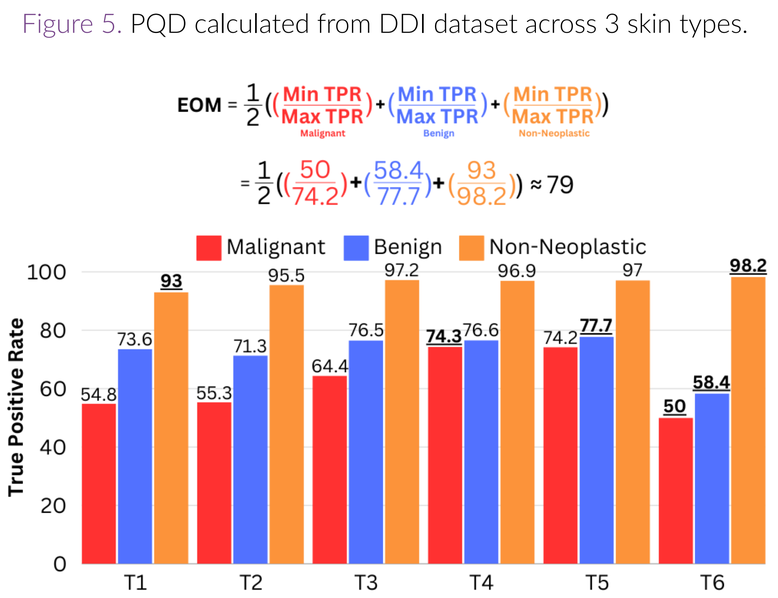


Figure 6. EOM calculated from Fitzpatrick17k dataset across 6 skin types, for PatchAlign.

Results and Discussion

Table 1. In-Domain classification on DDI Dataset

Model	Accuracy (%) ± Std-dev.				Fairness (%)	
	Avg	T12	T34	T56	*PQD*	*EOM*
BASE	82.4 ± 1.5	83.3 ± 1.0	74.6 ± 5.7	89.7 ± 2.2	77.0 ± 1.9	58.7 ± 4.3
FairDisCo	83.8 ± 0.4	88.6 ± 0.1	71.7 ± 2.2	92.0 ± 2.8	78.0 ± 4.5	63.7 ± 3.5
PatchAlign	**87.4 ± 1.2**	**89.6 ± 2.6**	**80.3 ± 5.7**	**92.3 ± 1.3**	**86.9 ± 6.1**	**69.6 ± 1.7**

Table 2. In-Domain Classification on Fitzpatrick17k Dataset

Model	Accuracy (%) ↑							Fairness (%)	
	Avg	T1	T2	T3	T4	T5	T6	PQD	EOM
BASE(ResNet-18)	85.0	82.6	81.7	85.0	88.8	90.7	82.8	88.9	64.0
FairDisCo	85.1	82.2	82.1	86.2	89.4	90.0	83.2	**90.2**	63.8
PatchAlign	**88.6**	82.3	**85.6**	**89.2**	**91.4**	**91.3**	**91.4**	90.0	**74.8**
BASE(ViT)	86.0	**83.5**	82.9	86.9	90.5	90.5	84.6	89.4	59.2
BASE(ViT)+GOT	86.6	81.8	84.2	88.4	91.4	91.3	83.3	89.4	59.2

Table 3. Out-Domain Classification on the Fitzpatrick17k Dataset

	Model	Accuracy							Fairness (%)	
		Avg	T1	T2	T3	T4	T5	T6	PQD	EOM
A	FairDisco	79.5	-	-	80.2	79.4	78.1	79.0	96.9	**71.2**
	PatchAlign	**84.0**	-	-	**84.2**	**84.0**	**83.1**	**84.6**	**98.3**	68.2
B	FairDisco	78.3	73.4	77.9	-	-	86.7	83.7	84.6	64.6
	PatchAlign	**82.4**	**78.6**	**82.1**	-	-	**89.5**	**86.1**	**87.8**	**75.1**
C	FairDisco	71.5	65.3	68.9	73.1	80.4	-	-	81.2	77.5
	PatchAlign	**77.6**	**74.4**	**75.6**	**78.2**	**83.5**	-	-	**89.1**	**78.1**

Want to know more?

- Scan to go to Github repository with code and link to paper.
- More evaluations further demonstrating accuracy and robustness in in-domain and out-domain classification.
- Ablation studies and comparison with Multi-Task Learning.

SCAN ME

gagan@iitbhilai.ac.in